Accent Conversion using Pre-trained Model and Synthesized Data from Voice Conversion

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Citations: : Nguyen, T.N., Pham, N.-Q., Waibel, A. (2022) Accent Conversion using Pre-trained Model and Synthesized Data from Voice Conversion. Proc. Interspeech 2022, 2583-2587, doi: 10.21437/Interspeech.2022-10729

2. Task

Accent conversion (specifically, reference-free end-to-end accent conversion).

3. Parallel data: required, beneficial, or not used?

`requires-parallel`

The proposed seq2seq architecture inherently requires parallel input-output pairs for training. Because natural parallel data (same voice, same content, different accent) does not exist, the study introduces a two-step pipeline to artificially synthesize a parallel corpus via Voice Conversion (VC) before training the AC model.

4. If parallel: what alignment method?

Attention/seq2seq implicit alignment. The models rely on

cross-attention modules (in FragmentVC) and location-sensitive attention mechanisms (in the Tacotron-style decoder) to align source features to target features with similar phonetic content.

5. Stated data bottleneck

“The main challenge in AC is that ground-truth data for the desired output is not available… There has not been a parallel corpus that contains pairs of audios having the same contents yet coming from the same speakers in different accents, the authors hence work on a solution to synthesize one as training input.”

6. Did they use parallel data for evaluation even when training non-parallel?

Yes (implicitly via synthesis). The entire methodology exists to generate synthetic parallel pairs to serve as ground-truth for both training and objective evaluation (comparing original vs. converted outputs).

7. Language(s) and data scale

English. Combines CMU-ARCTIC and L2-ARCTIC. 8 total speakers (4 native US, 4 Indian-accented), 1152 sentences per speaker. Synthesized dataset yielded ~9 hours of audio (approx. 9,000 synthetic parallel pairs).

8. Accent conversion specifically: what was the source–target pairing?

L2 ↔ L1 (Indian-accented to American-accented, and explicitly tested American-accented to Indian-accented to generate synthetic non-native training data).

9. Evaluation metrics used
Word Error Rate (WER via ASR), Accent Classifier Accuracy (ACC), and Subjective MOS (Accentedness and Speaker Similarity).

10. Tag
`context-cite` *(Note for paper: Use to argue that the lack of natural parallel speech data is so restrictive that researchers are forced to hallucinate/synthesize parallel data using complex VC pipelines just to train baseline AC models. Your corpus removes this need.)*

Can We Achieve High-quality Direct Speech-to-Speech Translation without Parallel Speech Data?

Cite

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@inproceedings{fang2024can,
title={Can we achieve high-quality direct speech-to-speech translation without parallel speech data?},
author={Fang, Qingkai and Zhang, Shaolei and Ma, Zhengrui and Zhang, Min and Feng, Yang},
booktitle={Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)},
pages={7264–7277},
year={2024}
}
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2. Task

Direct Speech-to-Speech Translation (S2ST) — specifically, proposing a zero-shot model (ComSpeech-ZS) built from composite sub-models.

3. Parallel data: required, beneficial, or not used?

`non-parallel` (The core contribution is achieving high-quality S2ST *without* parallel speech data by connecting independent pretrained models).

(Note for your paper: This is a perfect example of a system that explicitly avoids the parallel data requirement, which you must acknowledge in §2.6 to keep your review balanced).

4. If parallel: what alignment method?

The model *replaces* the need for parallel S2ST data by using a Connectionist Temporal Classification (CTC) based vocabulary adaptor. It aligns the subword text representation space of a Speech-to-Text Translation (S2TT) model with the phoneme/character input space of a Text-to-Speech (TTS) model via contrastive learning and forced alignment.

5. Stated data bottleneck

“However, the training of these models still requires large-scale parallel speech data comprising <source speech, target text, target speech> triplets, which is extremely challenging to collect.”

6. Did they use parallel data for evaluation even when training non-parallel?

Yes. While the model trains without parallel data (using only S2TT and TTS datasets), the evaluation on the CVSS dataset directly measures against parallel references (using ASR-BLEU to compare translated speech to ground-truth text, and BLASER 2.0 to evaluate cross-lingual semantic similarity against reference speech).

7. Language(s) and data scale

Evaluated on French/German/Spanish → English.
Trained on the CVSS dataset.
The zero-shot scenario used 264h (Fr), 184h (De), and 113h

(Es) of S2TT data, and 191h of non-overlapping English TTS data.

8. Accent conversion specifically: what was the source–target pairing?
Not applicable (this is Speech-to-Speech cross-lingual translation, not intra-lingual Accent Conversion).

9. Evaluation metrics used
ASR-BLEU (transcribing the generated speech and computing BLEU against text references) and BLASER 2.0 (evaluating cross-lingual semantic similarity directly on the speech).

10. Tag
`context-cite` *(Note for paper: Use this to show that while state-of-the-art architectures are desperately trying to bypass the "extremely challenging to collect" parallel data via complex adapter mechanisms (CTC + contrastive learning), parallel data remains the undisputed necessity for their baseline evaluations and supervised upper-bounds).*

CVSS Corpus and Massively Multilingual Speech-to-Speech Translation

Citation:

@inproceedings{jia2022cvss,
title={CVSS corpus and massively multilingual speech-to-speech translation},
author={Jia, Ye and Ramanovich, Michelle Tadmor and Wang, Quan and Zen, Heiga},
booktitle={Proceedings of the thirteenth language resources and evaluation conference},
pages={6691–6703},
year={2022}
}

2. Task

Speech-to-Speech Translation (S2ST) Corpus Creation and Baseline Evaluation.

3. Parallel data: required, beneficial, or not used?

`benefits-from-parallel` (The paper *introduces* a massive parallel corpus because direct/cascade S2ST models benefit immensely from it, though modern systems try to avoid the cost).

4. If parallel: what alignment method?

Not applicable in the traditional sense, as this paper describes the *creation* of the corpus. The alignment is

inherent in the dataset design: synthetic target speech was generated from normalized translated text using a TTS model (PnG NAT) to create sentence-level parallel pairs.

5. Stated data bottleneck

“However, as of today, publicly available corpora directly suitable for such research [direct S2ST] are extremely limited… Collecting a large amount of synthetic speech is of much lower cost than collecting human recordings, in absence of a directly usable S2ST corpus.”

6. Did they use parallel data for evaluation even when training non-parallel?

Yes. The CVSS corpus itself is designed to be the parallel evaluation ground-truth. They evaluate translation quality by computing BLEU on the ASR transcription of the generated target speech against the normalized reference translation text.

7. Language(s) and data scale

Massively Multilingual. 21 source languages to English.
Scale: 1,872 hours of total speech for CVSS-C (single voice) and 1,937 hours for CVSS-T (voice transferred).

8. Accent conversion specifically: what was the source–target pairing?

Not applicable (this is Speech-to-Speech translation, not accent conversion). However, the CVSS-T dataset transfers

the *voice* (speaker identity) from the source language to the target language, creating a pseudo-parallel corpus for voice-preserving S2ST.

9. Evaluation metrics used

ASR-BLEU (for translation quality) and Subjective MOS (for naturalness and speaker similarity).

10. Tag

`context-cite` *(Note for paper: Use this to illustrate the scale required for modern speech tasks. While CVSS uses synthetic target speech to achieve its massive scale, your BPAC corpus provides natural, human-recorded parallel data, which is much harder to collect but essential for capturing genuine phonetic nuance).*

DeepConversion: Voice conversion with limited parallel training data

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**DOI: ** 10.1016/j.specom.2020.05.004 / 2020 /

**Citation: **

@article{zhang2020deepconversion,
title={Deepconversion: Voice conversion with limited parallel training data},
author={Zhang, Mingyang and Sisman, Berrak and Zhao, Li and Li, Haizhou},
journal={Speech Communication},
volume={122},
pages={31–43},
year={2020},
publisher={Elsevier}
}

2. Task

Voice Conversion (specifically, optimizing spectral mapping and vocoding under limited data conditions).

3. Parallel data: required, beneficial, or not used?

`benefits-from-parallel` (The paper explicitly uses parallel data to fine-tune and adapt a general model, proving that while non-parallel data builds the foundation, parallel data is necessary for high-fidelity conversion).

4. If parallel: what alignment method?

Dynamic Time Warping (DTW) frame alignment. The source and target signals (which share the same content but have

different temporal dynamics) are aligned using DTW to form source-target acoustic feature pairs before adaptation.

5. Stated data bottleneck

“A deep neural network approach to voice conversion usually depends on a large amount of parallel training data from source and target speakers… In voice conversion, we typically use a small amount of parallel or non-parallel training data for practical reasons, that is much smaller than that in speech synthesis.”

6. Did they use parallel data for evaluation even when training non-parallel?

Yes. To compute the objective Mel-Cepstral Distortion (MCD) metric, parallel pairs of generated MCEP and ground-truth target MCEP were required.

7. Language(s) and data scale

English (CMU ARCTIC database).

Scale: The “large” non-parallel dataset consisted of 4 speakers (~4,500 utterances total). The “limited parallel data” used for adaptation consisted of only 50 to 200 utterances (a few minutes) between the specific source-target pairs (CLB, SLT, BDL).

8. Accent conversion specifically: what was the source–target pairing?

Speaker-to-speaker (inter-gender and intra-gender among female/male speakers).

9. Evaluation metrics used

Mel-Cepstral Distortion (MCD) for objective evaluation. Subjective evaluations included AB preference tests, ABX preference tests, Best Worst Scaling (BWS), and Mean Opinion Scores (MOS) for speech quality and naturalness.

10. Tag

`context-cite` *(Note for paper: Use this to demonstrate the hybrid approach: while large-scale non-parallel data can bootstrap general phonetic models (via PPGs), parallel data remains crucial for the "fine-grained" error reduction and final acoustic mapping that produces high-quality voice conversion. BPAC provides exactly this kind of precise, fine-grained parallel data).*

Invertible Voice Conversion with Parallel Data

**Citations: **

@inproceedings{cai2024invertible,
title={Invertible voice conversion with parallel data},
author={Cai, Zexin and Li, Ming},
booktitle={ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)},
pages={10041–10045},
year={2024},
organization={IEEE}
}

2. Task

Voice Conversion / Audio Security (Anti-spoofing).

3. Parallel data: required, beneficial, or not used?

`requires-parallel` (The proposed invertible architecture relies on paired parallel data to learn the exact bidirectional mapping between source and target).

4. If parallel: what alignment method?

Dynamic Time Warping (DTW) is used to align the acoustic features (Mel-spectrograms) for the parallel pairs during the data preparation stage.

5. Stated data bottleneck

“It’s important to note that, up to this point, achieving invertibility has necessitated the use of parallel data, which can be associated with significant costs. Our future

endeavors will primarily revolve around the goal of achieving invertibility at the waveform level, using non-parallel data, with the aim of enhancing the robustness and cost-effectiveness of our approach."

6. Did they use parallel data for evaluation even when training non-parallel?

Not applicable, as the training itself requires parallel data. However, the evaluation fundamentally relies on the parallel nature of the test set to compute objective metrics (like Mel-Spectrogram Distortion, MSD) between the synthesized outputs and the ground-truth target utterances.

7. Language(s) and data scale

English (CMU ARCTIC corpus).
Scale: 4 speakers ('bdl', 'slt', 'clb', 'rms'). 1,000 utterances per speaker for training (totaling 4,000 parallel utterances across various pairings).

8. Accent conversion specifically: what was the source–target pairing?

Speaker-to-speaker (inter-gender and intra-gender among the four selected speakers).

9. Evaluation metrics used

Mel-Spectrogram Distortion (MSD) for objective performance. Subjective Mean Opinion Scores (MOS) with a

95% confidence interval for Naturalness and Speaker Similarity.

10. Tag

`context-cite` *(Note for paper: Use this to emphasize that advanced capabilities in VC—such as exact mathematical invertibility for forensic audio tracing and anti-spoofing—are currently only possible through the use of parallel speech data. This positions BPAC as the exact kind of resource needed to extend these high-security features to regional dialects).*

Parallel and Limited Data Voice Conversion Using Stochastic Variational Deep Kernel Learning

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citations:

@article{jafaryani2022parallel,
title={Parallel voice conversion with limited training data using stochastic variational deep kernel learning},
author={Jafaryani, Mohamadreza and Sheikhzadeh, Hamid and Pourahmadi, Vahid},

journal={Engineering Applications of Artificial Intelligence},
volume={115},
pages={105279},
year={2022},
publisher={Elsevier}
}

2. Task
Voice Conversion (specifically, spectral mapping with limited parallel training data).

3. Parallel data: required, beneficial, or not used?
`requires-parallel` (The proposed method operates strictly in a frame-level mapping paradigm that relies on dynamic time warping to align source and target acoustic features).

4. If parallel: what alignment method?
Dynamic Time Warping (DTW) frame alignment. It aligns 24 Mel-Cepstral Coefficients (MCCs) extracted from the spectral envelopes of the source and target utterances.

5. Stated data bottleneck
“Typically, voice conversion is regarded as an engineering problem with limited training data. The reliance on massive amounts of data hinders the practical applicability of deep learning approaches… On the other hand, statistical methods are effective with limited data but have difficulties in modelling complex mapping functions.”

6. Did they use parallel data for evaluation even when training non-parallel?

Not applicable, as the training itself uses parallel data. Evaluation utilizes this parallel testing data to compute the objective Mel Cepstral Distortion (MCD) metric between the converted speech and the target reference.

7. Language(s) and data scale

English (CMU ARCTIC dataset).
Scale: *Extremely limited*. The training data consisted of only 20 sentences per speaker pair, equating to approximately 80 seconds of audio. Validation used 10 sentences, and testing used 20 sentences.

8. Accent conversion specifically: what was the source–target pairing?

Speaker-to-speaker (intergender and intragender conversion among four speakers: BDL, RMS, CLB, SLT).

9. Evaluation metrics used

Objective: Mel Cepstral Distortion (MCD).
Subjective: Mean Opinion Score (MOS), Quality preference test, and Similarity preference test.

10. Tag

`context-cite` *(Note for paper: Use this to underscore the persistent tension in the field: traditional Deep Neural Networks overfit on small datasets, while statistical methods underfit*

complex functions. While SVDKL mitigates this algorithmically for 80-second subsets, a robust, large-scale parallel corpus like BPAC fundamentally solves the "expressive capability vs. overfitting" dilemma at the data level).